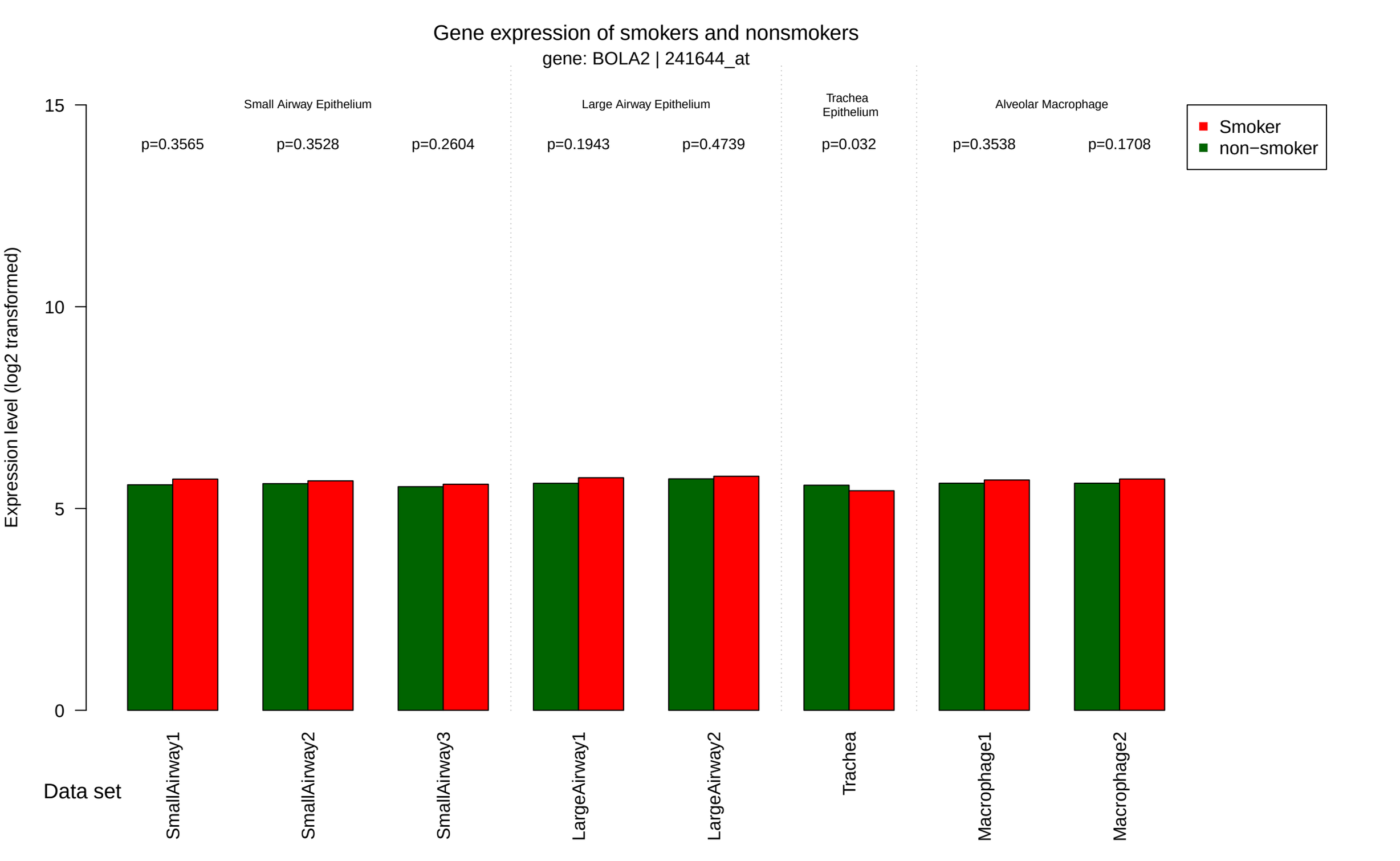
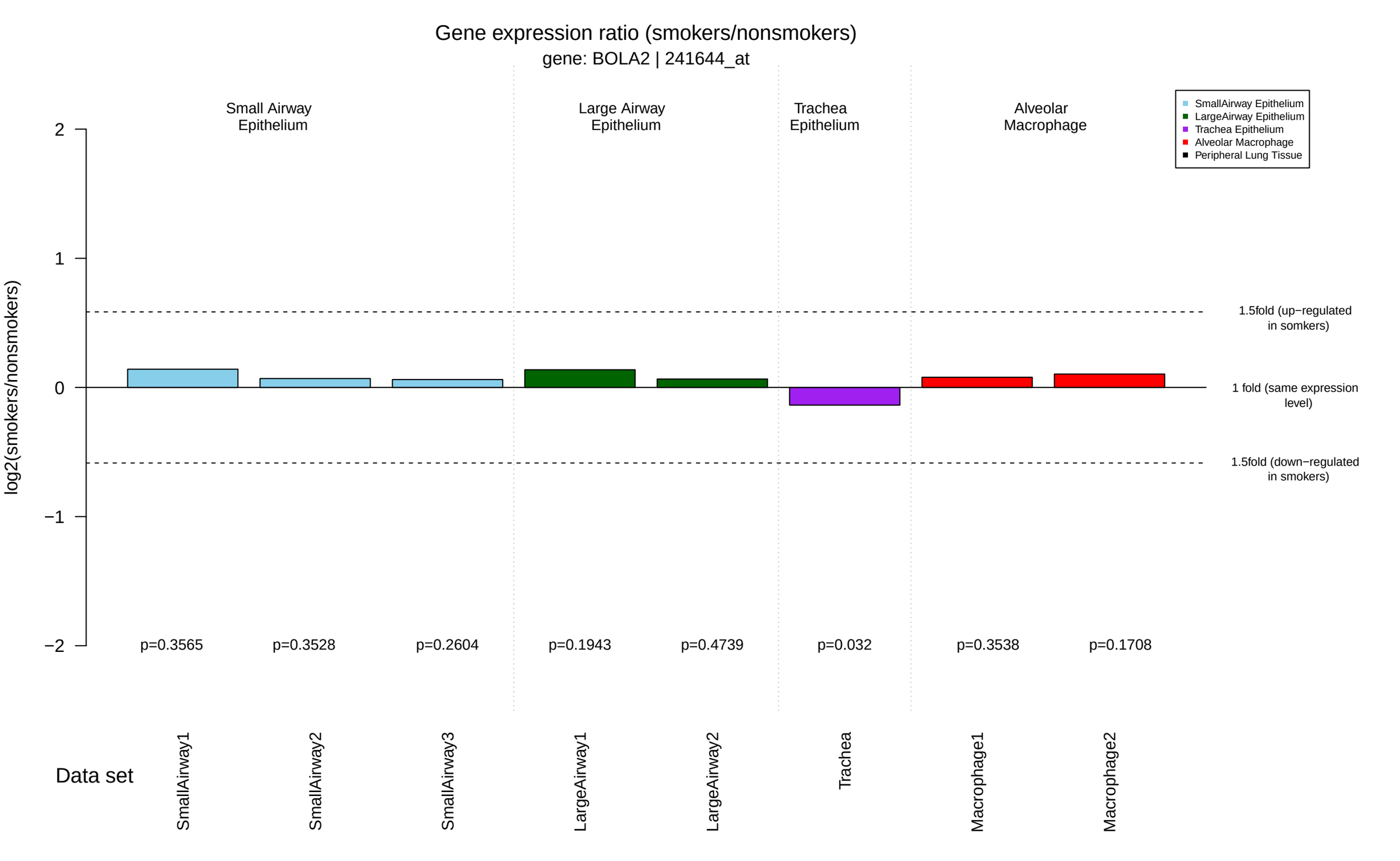
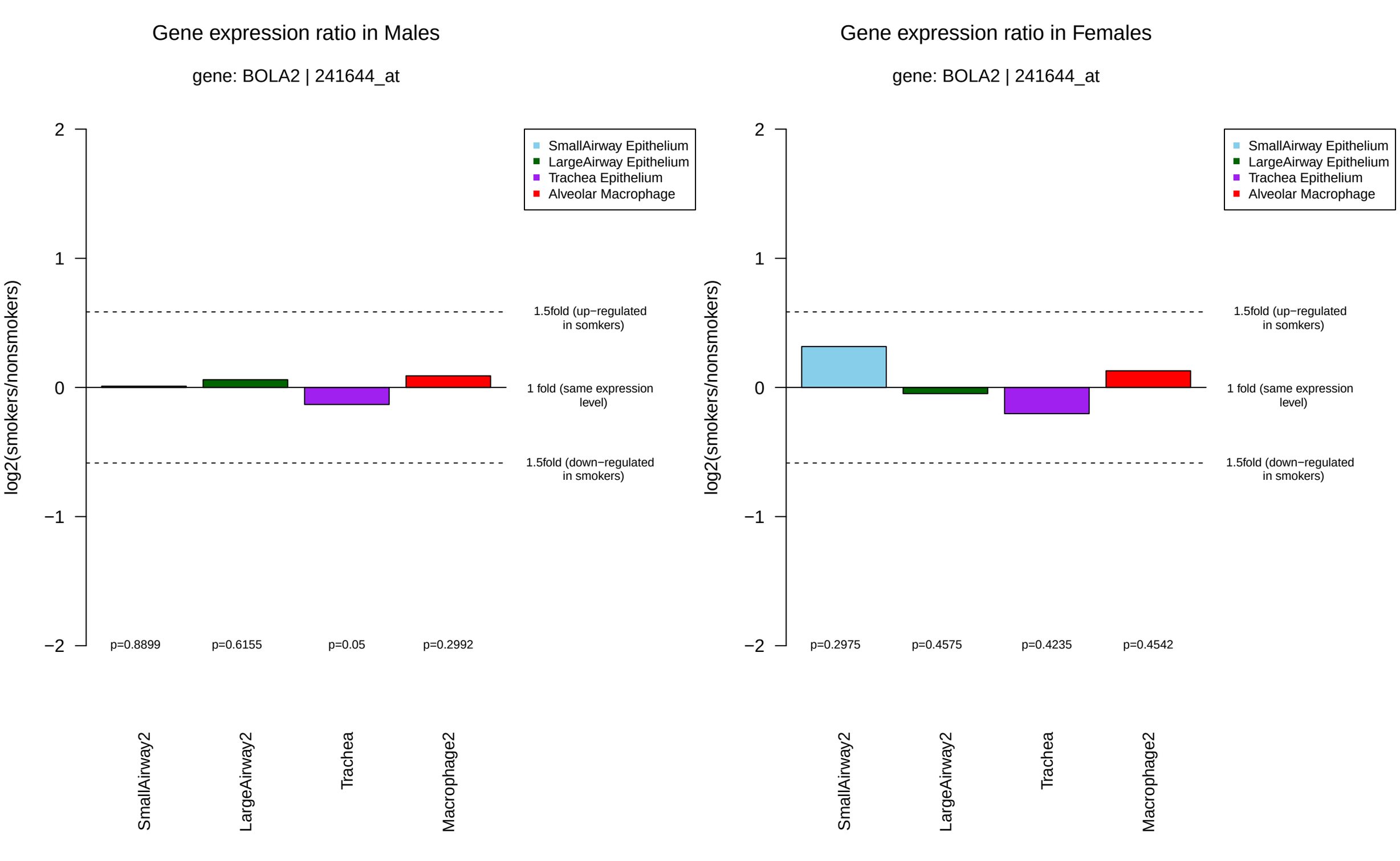
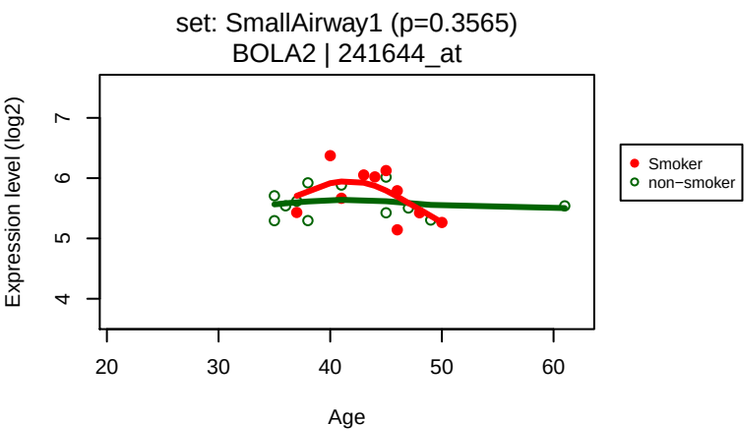
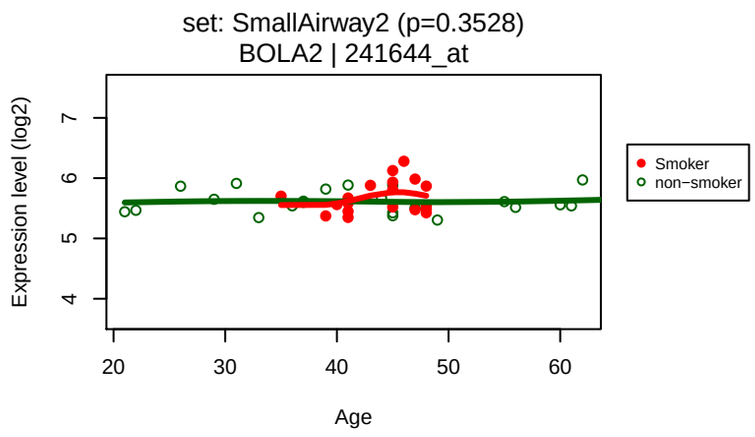
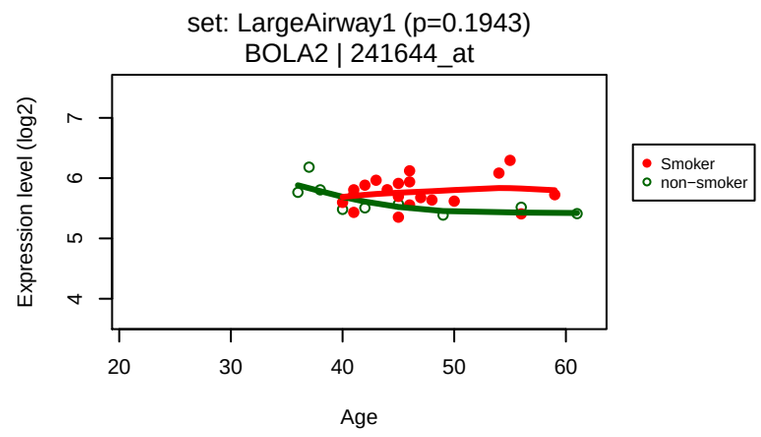

Gene expression of smokers and nonsmokers
gene: BOLA2 | 241644_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3565
p=0.3528
p=0.2604
p=0.1943
p=0.4739
p=0.032
p=0.3538
p=0.1708
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

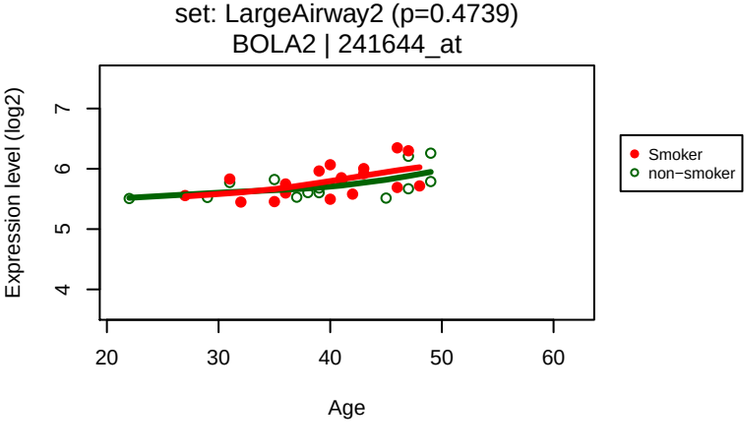

Gene expression ratio (smokers/nonsmokers)
gene: BOLA2 | 241644_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3565
p=0.3528
p=0.2604
p=0.1943
p=0.4739
p=0.032
p=0.3538
p=0.1708
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

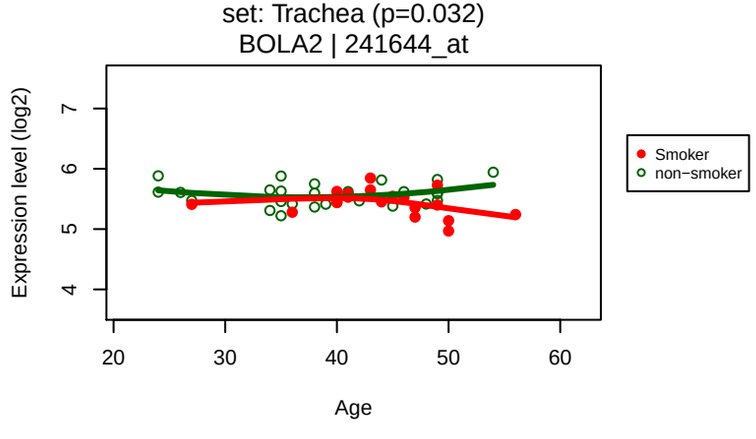

Gene expression ratio in Males
gene: BOLA2 | 241644_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8899
p=0.6155
p=0.05
p=0.2992
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: BOLA2 | 241644_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2975
p=0.4575
p=0.4235
p=0.4542
SmallAirway2
LargeAirway2
Trachea
Macrophage2

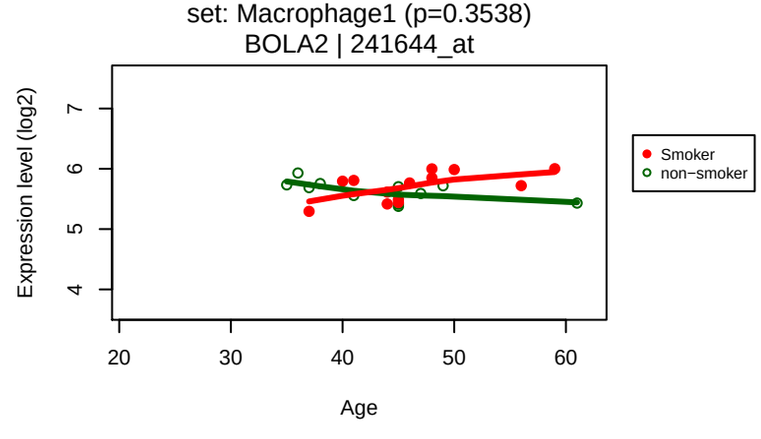

set: SmallAirway1 (p=0.3565)
BOLA2 | 241644_at
Expression level (log2)
Age
Smoker
non−smoker

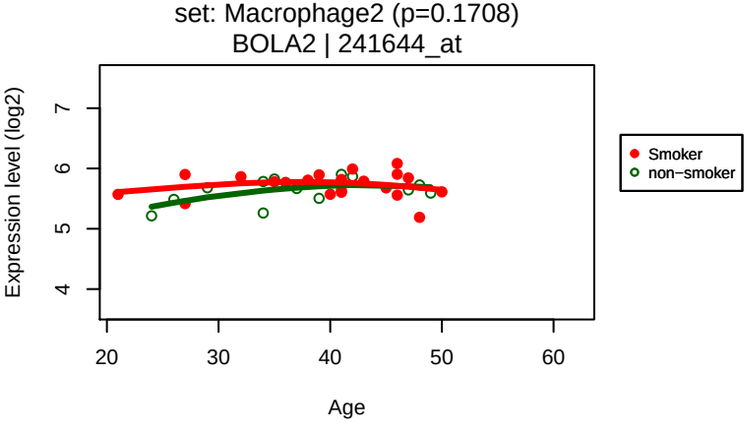

set: SmallAirway2 (p=0.3528)
BOLA2 | 241644_at
Expression level (log2)
Age
Smoker
non−smoker

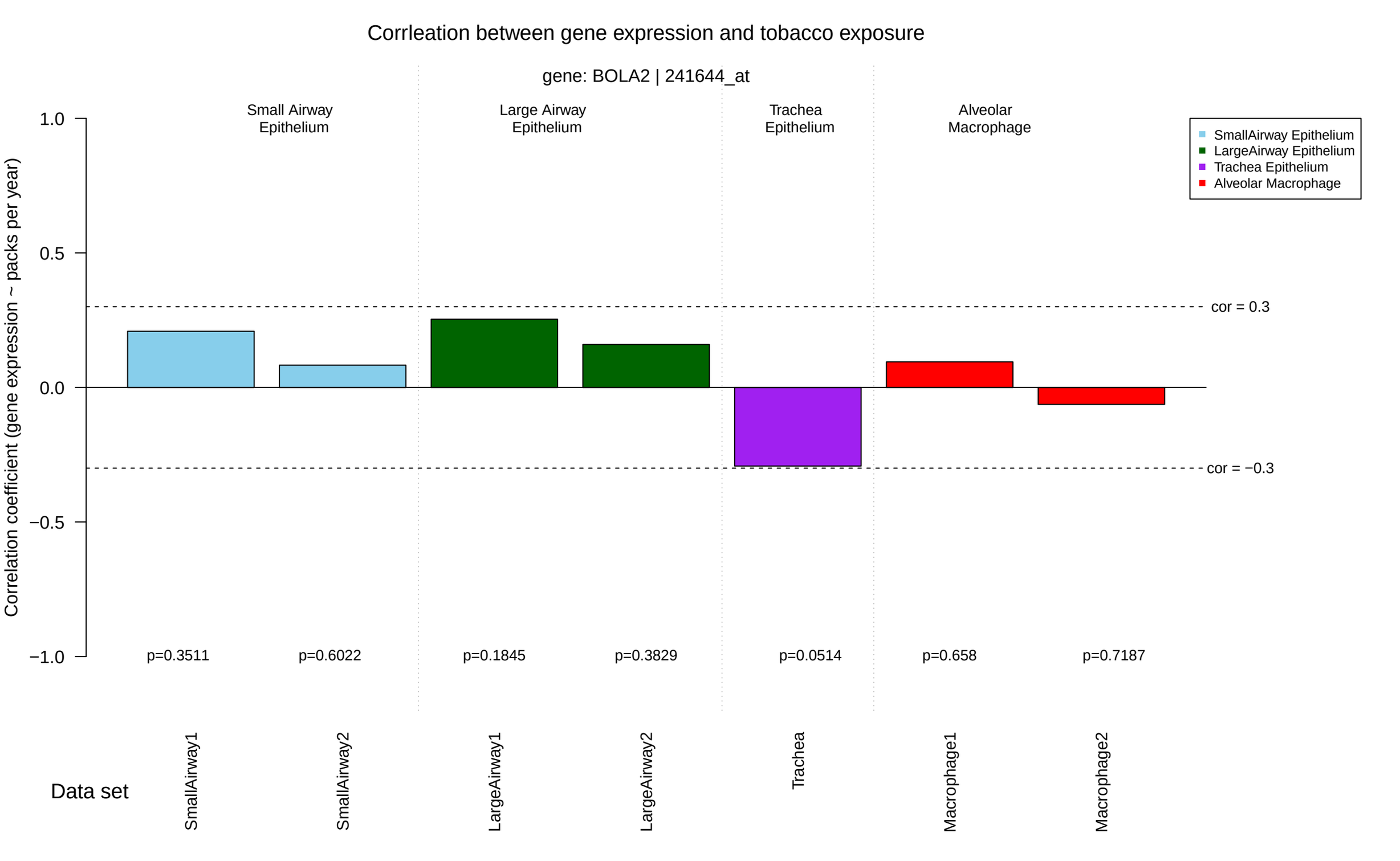

set: LargeAirway1 (p=0.1943)
BOLA2 | 241644_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4739)
BOLA2 | 241644_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.032)
BOLA2 | 241644_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3538)
BOLA2 | 241644_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1708)
BOLA2 | 241644_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: BOLA2 | 241644_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3511
p=0.6022
p=0.1845
p=0.3829
p=0.0514
p=0.658
p=0.7187
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2